

Homework I

Paul Haskins
22338905

February 17, 2026

Contents

0.1	Question 1	1
0.2	Question 2	1
0.3	Question 3	1
0.4	Question 4	1

0.1 Question 1

Two identical spaceships, each of proper length L , approach Earth with equal speeds (as measured by an observer on Earth) but from opposite directions. One spaceships captain measures the length of the other to be ℓ . What is the speed of each spaceship as measured by the observer on Earth?

Call one spaceship A and the other spaceship B. If spaceship A measures the distance to spaceship B to be ℓ and each are a distance of the 'proper length L ' from earth. The the distance that spaceship A measures to the earth will be $\frac{\ell}{2}$. In the frame of the earth this distance is $L = \frac{\ell}{2\gamma}$

$$L = \frac{\ell}{2} \sqrt{1 - \frac{\vec{v}^2}{c^2}}$$

$$|v| = \sqrt{\left[1 - \left(\frac{2L}{\ell}\right)^2\right]} c$$

$$\vec{v} = -c \sqrt{\left[1 - \left(\frac{2L}{\ell}\right)^2\right]} \hat{r}.$$

where we have defined $\hat{r}$ to be the radial distance from the center of the earth.

0.2 Question 2

0.3 Question 3

0.4 Question 4

Given

$$S = \frac{1}{16\pi G_N} \int d^n x \sqrt{-g} (R - 2\Lambda) \quad (1)$$

(a) Derive the equations of motion.

]

$$\delta S = \frac{1}{16\pi G_N} \int d^n x \delta [\sqrt{-g} (g^{\mu\nu} R_{\mu\nu} - 2\Lambda)]$$

$$\delta S = \frac{1}{16\pi G_N} \int d^n x [\sqrt{-g} g^{\mu\nu} (\delta R_{\mu\nu}) + \sqrt{-g} R_{\mu\nu} (\delta g^{\mu\nu}) + (R - 2\Lambda) (\delta \sqrt{-g})].$$

Look at the first term

$$R_{\mu\nu} \equiv R_{\mu\rho\nu}^{\rho}$$

$$R_{\mu\rho\nu}^{\sigma} \equiv \partial_{\rho} \Gamma_{\mu\nu}^{\sigma} + \Gamma_{\rho\lambda}^{\sigma} \Gamma_{\nu\mu}^{\lambda} - (\rho \leftrightarrow \nu).$$

Explain below

$$\nabla_{\rho} (\delta \Gamma_{\nu\mu}^{\sigma}) = \Gamma_{\nu\mu,\rho}^{\sigma} + \Gamma_{\rho\lambda}^{\sigma} \delta \Gamma_{\nu\mu}^{\lambda} [-\Gamma_{\rho\nu}^{\lambda} \delta \Gamma_{\lambda\mu}^{\sigma} - \Gamma_{\rho\mu}^{\lambda} \delta \Gamma_{\nu\lambda}^{\sigma}].$$

Now if we look at the second term

Now if we look at the third term

Notice

$$\sqrt{-g} = \sqrt{-e^{Tr(\ln(g_{\mu\nu}))}} = \sqrt{-e^{-Tr(\ln(g^{\mu\nu}))}}$$

$$\delta \sqrt{-g} = \delta \left(-e^{-\frac{1}{2} Tr(\ln(g^{\mu\nu}))} \right) = \left(-e^{-\frac{1}{2} Tr(\ln(g^{\mu\nu}))} \right) \cdot \delta \left(-\frac{1}{2} Tr(\ln(g^{\mu\nu})) \right)$$

$$= -\frac{1}{2} \sqrt{-g} \delta (Tr(\ln g^{\mu\nu})).$$

We can use $\delta \ln A = A^{-1} \delta A$ and $\delta Tr(A) = Tr(\delta A)$

$$\delta \sqrt{-g} = -\frac{1}{2} \sqrt{-g} Tr(g_{\mu\nu} \delta g^{\mu\nu}).$$

$$\begin{aligned} \delta S &= \frac{1}{16\pi G_N} \int d^n x [\sqrt{-g} R_{\mu\nu} (\delta g^{\mu\nu}) + (R - 2\Lambda) (\delta \sqrt{-g})] \\ &= \frac{1}{16\pi G_N} \int d^n x \left[\sqrt{-g} (R_{\mu\nu} \delta g^{\mu\nu}) + (R - 2\Lambda) \left(-\frac{1}{2} \sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu} \right) \right] \\ &= \frac{1}{16\pi G_N} \int d^n x \left[\sqrt{-g} R_{\mu\nu} + (R - 2\Lambda) \left(-\frac{1}{2} \sqrt{-g} g_{\mu\nu} \right) \right] \delta g^{\mu\nu} \\ &\implies \sqrt{-g} R_{\mu\nu} + (R - 2\Lambda) \left(-\frac{1}{2} \sqrt{-g} g_{\mu\nu} \right) = 0 \\ 0 &= R_{\mu\nu} - \frac{1}{2} (R - 2\Lambda) g_{\mu\nu} \\ \Lambda = 0 &\implies R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 0 \end{aligned}$$

(b) Consider 4-d Anti de Sitter metric

$$ds^2 = \frac{L^2}{z^2} (dz^2 + dx^2 + dy^2 - dt^2) \quad (2)$$

We need to compute the Ricci Tensor which has $\frac{n^2(n^2-1)}{12}$ independent components ($n = 4 \implies 20$ independent components).

To find the Ricci Tensor we have the following procedure

$$ds^2 \rightarrow g_{**} \rightarrow \Gamma_{**}^* \rightarrow R_{***}^* \quad (3)$$

We have $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$. So we can identify

$$g_{\mu\nu} = \frac{L^2}{z^2} \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (4)$$

$$\Gamma_{\mu\nu}^\rho = \frac{1}{2} g^{\rho\lambda} (\partial_\mu g_{\lambda\nu} + \partial_\nu g_{\mu\lambda} - \partial_\lambda g_{\mu\nu}) \quad (5)$$

$$R_{\mu\rho\nu}^\sigma = \partial_\rho \Gamma_{\mu\nu}^\sigma + \Gamma_{\rho\lambda}^\sigma \Gamma_{\nu\mu}^\lambda - (\rho \leftrightarrow \nu) \quad (6)$$